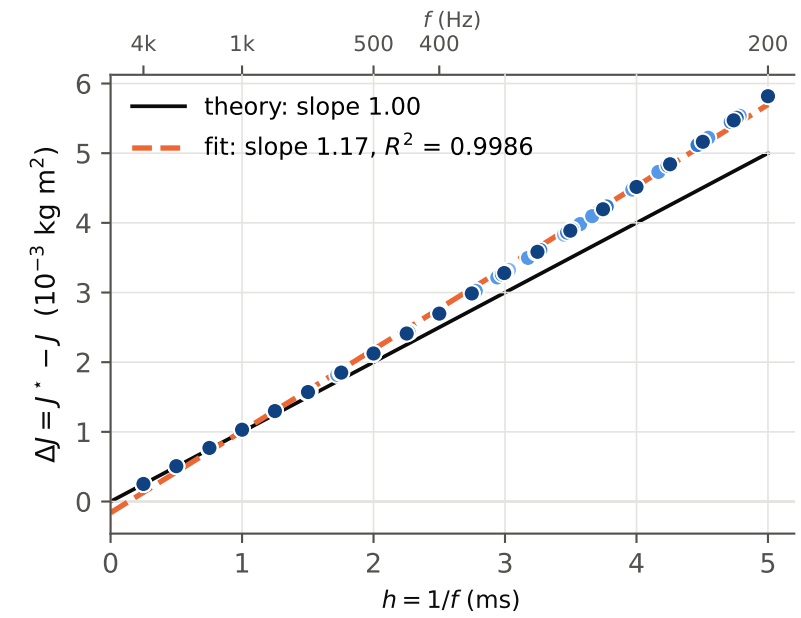
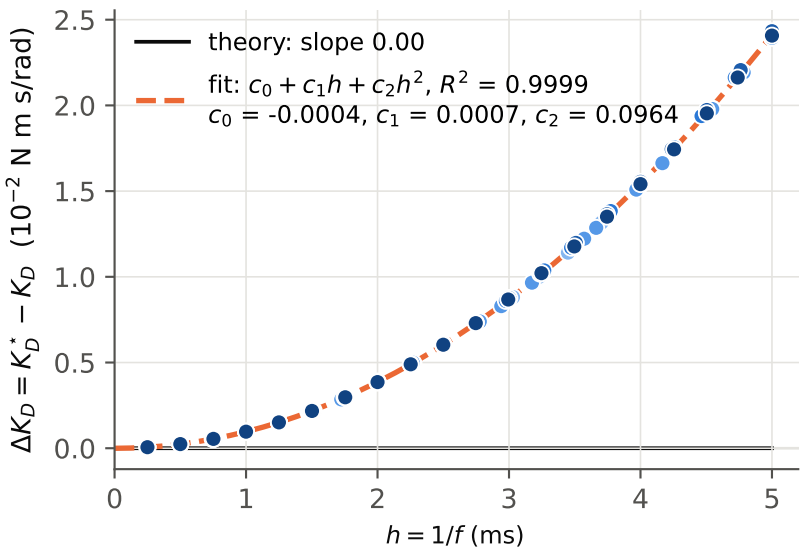
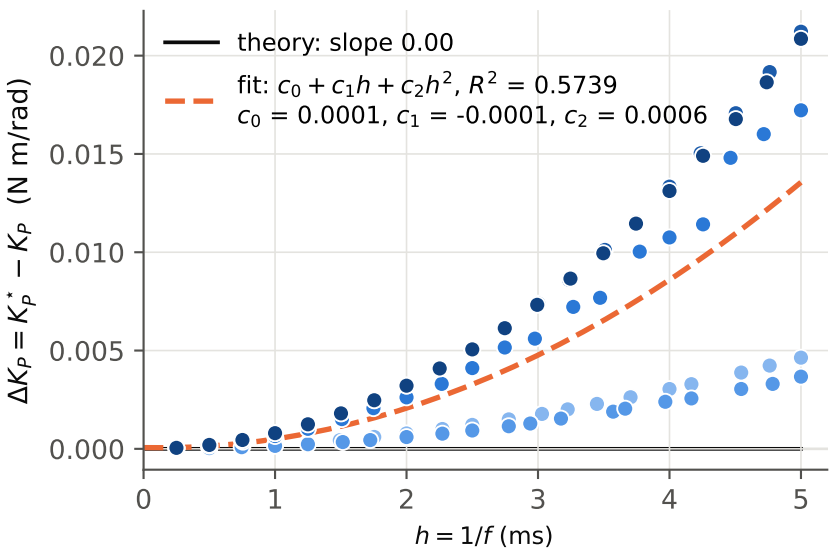
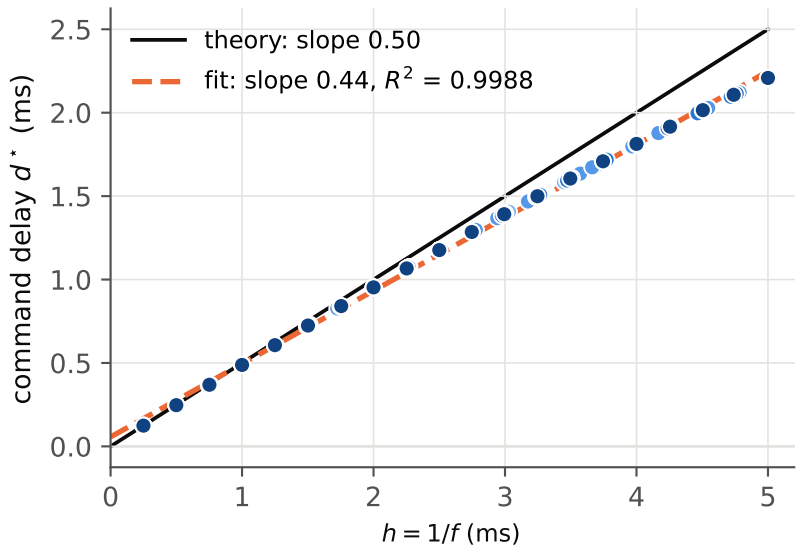
$f_{ctrl} \approx$ 10 Hz · $f_{ctrl} \approx$ 20 Hz · $f_{ctrl} \approx$ 50 Hz · $f_{ctrl} \approx$ 100 Hz · $f_{ctrl} \approx$ 200 Hz

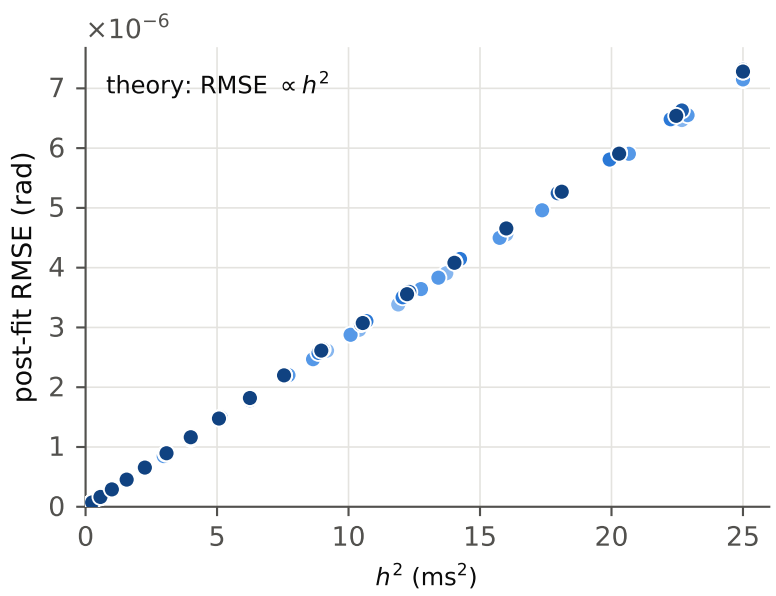
A: Euler + explicit PD
fitted: J, K_P, K_D, d



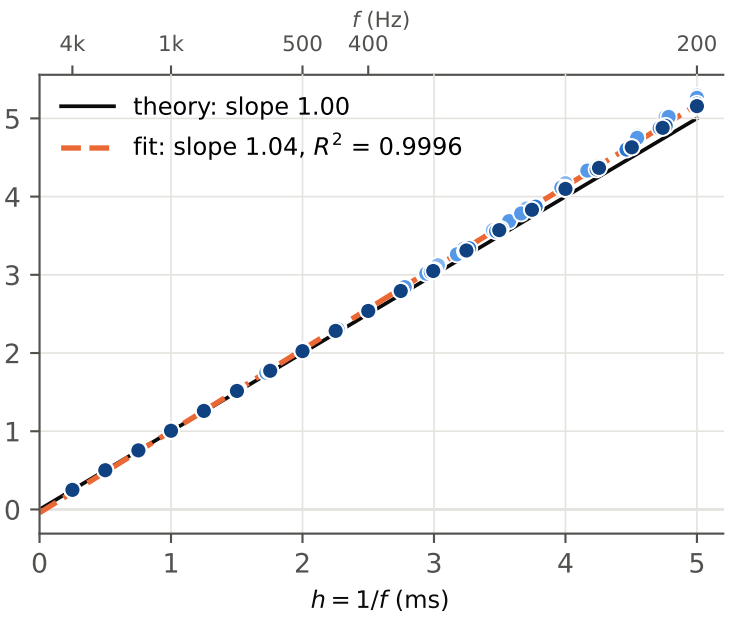


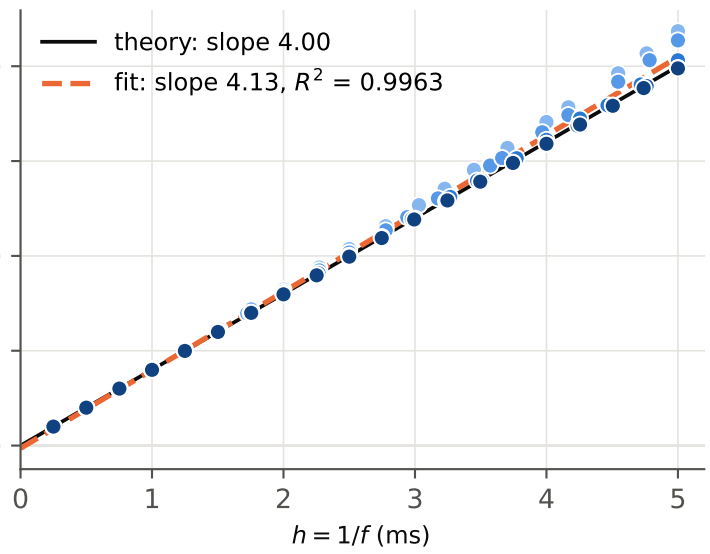


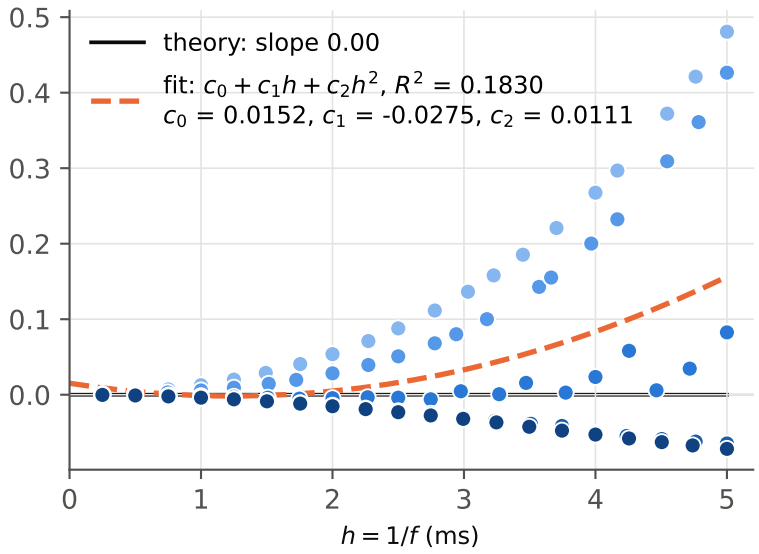




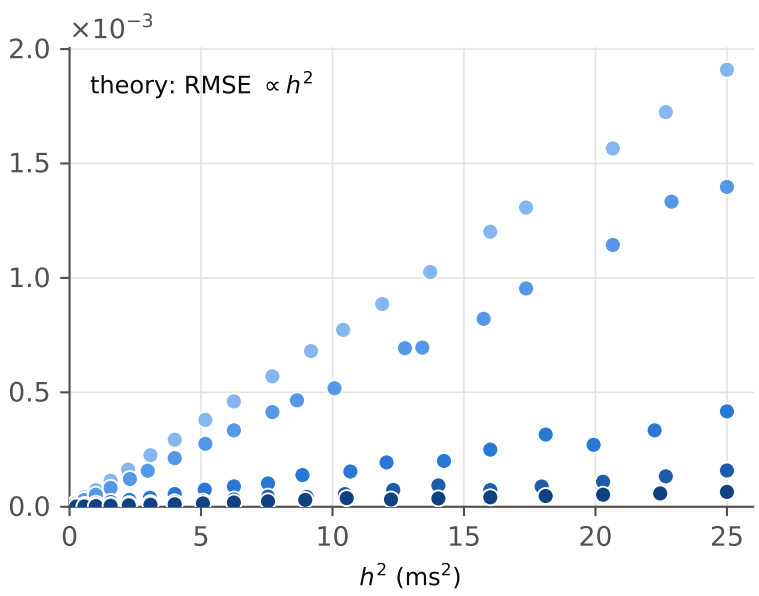
B: RK4 + explicit PD (torque held)
fitted: J, K_P, K_D



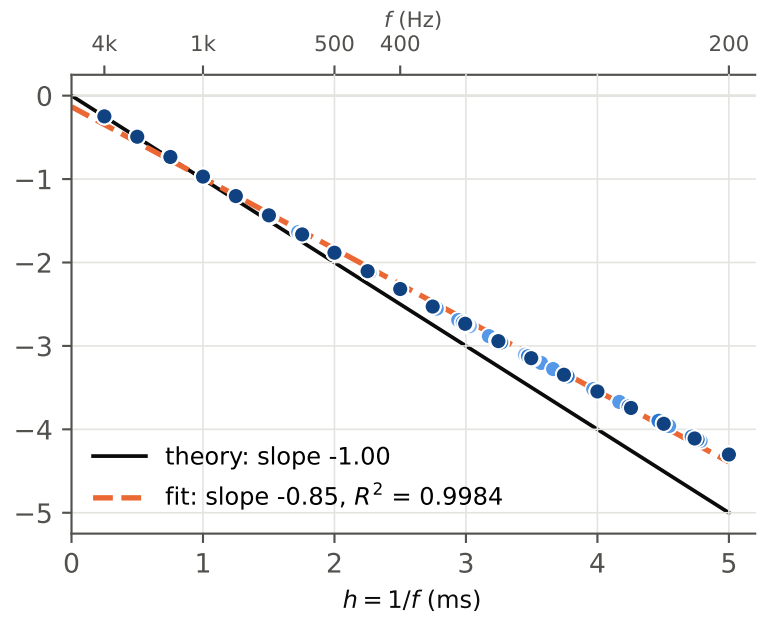


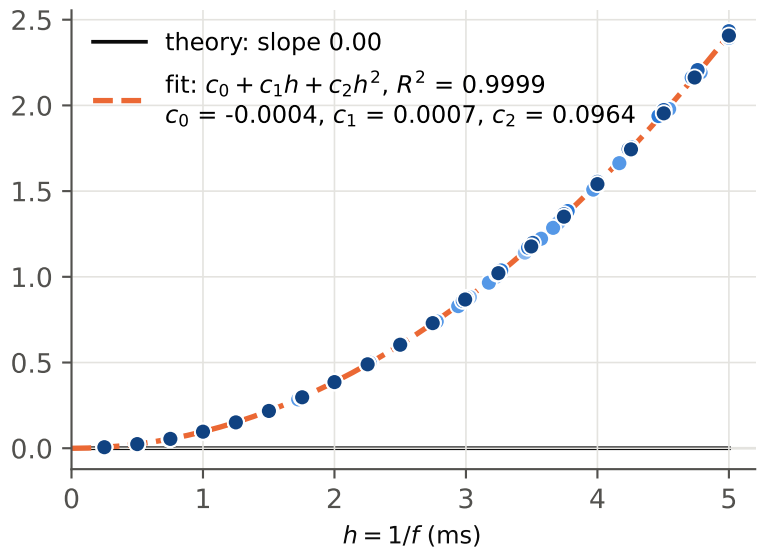


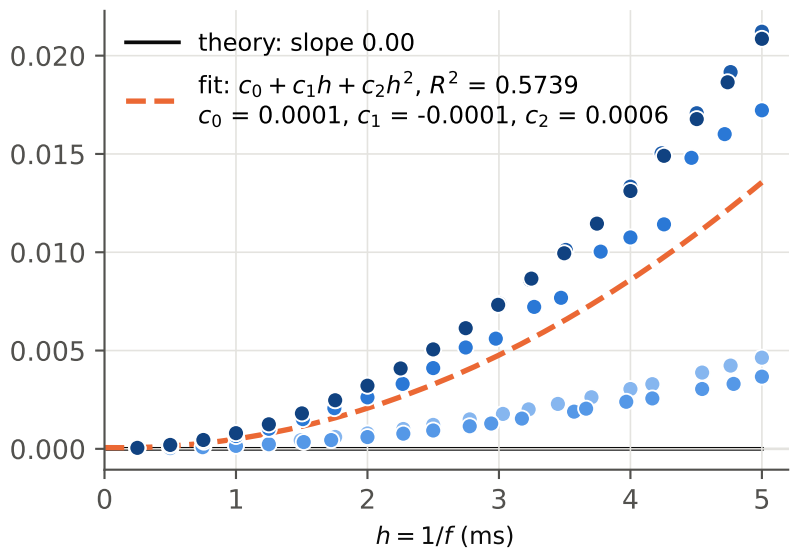
not fitted:
no command advance in this scheme

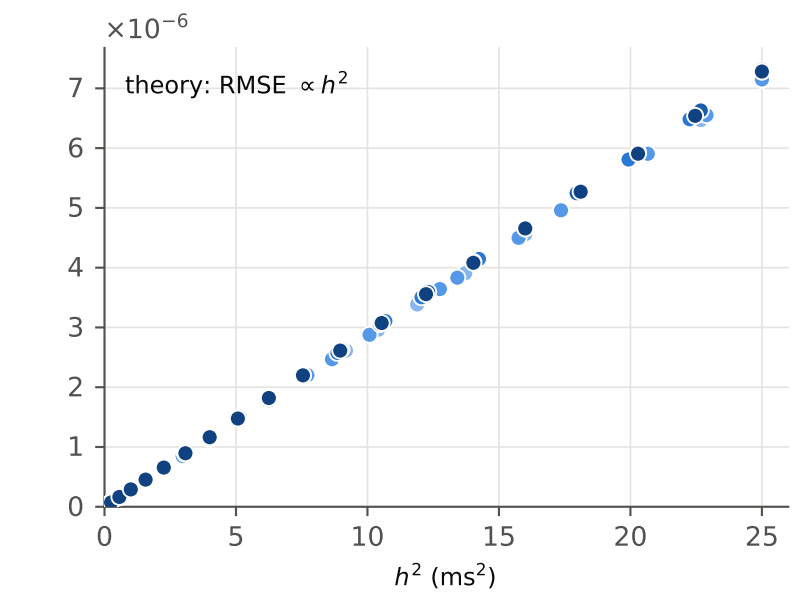


C: implicitfast + native PD
fitted: J, K_P, K_D, d









Target h = 0.25, 0.5, ..., 5 ms (f = 200-4000 Hz), realized with integer f_{ctrl} (Hz) and integer decimation, within 2.3% of the target.
Parameter rows: first-order theory (solid); fit (dashed), linear, or quadratic where the theory slope is 0. Last row: post-fit RMSE against h^2.